



GOA ELECTRONICS LIMITED

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Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	01-Mar-2021	Siya Chavan (Senior Lead Testing)	Anant Yende (CTO)	-
1.1	22-May-2023	Siya Chavan (Senior Lead Testing)	Anant Yende (CTO)	Following changes done 1. Testing Compliance Timeline Standard added. 2. Testing Standard tools added. 3. Sensitive personal data and Information security testing added. 4. Testing Phase Standards added. 5. Standard Checklist for testing Department workflow added. 6. Font and Style have been revised.
1.2	10-Jul-2024	Siya Chavan (Senior Lead Testing)	Anant Yende (CTO)	Following changes done 1. Testing clearance certificate added 2. Updated testing phase standard. 3. Changes in Validation Field Matrix
1.3	28-Sept-2024	Siya Chavan (Senior Lead Testing)	Anant Yende (CTO)	Following changes done 1. Updated Testing Applicability Standards 2. Updated Standard Testing Tool 3. Updated Standard Unit test scenarios for any application/ website 4. Updated Non-functional test scenarios for any application / website 5. Reordered and Added Annexure IV , V, VI , VII , VIII, IX
1.4	11-Feb-2025	Siya Chavan (Senior Lead Testing)	Anant Yende (CTO)	1. Updated Clearance Certificate document 2. Security testing changes
1.5	01-Aug-2025	Siya Chavan (Senior Lead Testing)	Anant Yende (CTO)	1. Added PM deviation comment clause for High Priority testing tickets (Section 4) 2. Added "On Hold" marking option for PM/TL in case of dependencies (Section 8)

REVISION HISTORY



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1. Introduction

This document defines the standard practices and procedures for software testing at Goa Electronics Limited. It aims to ensure the delivery of high-quality software products through consistent and effective testing processes.

- It is mandatory to get the application/system tested for any project/module from the testing team.
- The Testing will be carried out by the testing team as per the standard timelines defined in this standards and responses towards the test to be submitted by Technical Lead in standard notified time.
- The Testing Vertical Lead will decide test applicability based on the project/module under test, in addition to request for test submitted by Technical Lead.
- The module cannot be released to client till the Test Clearance Certificate is issued(Refer Annexure III).

2. Test Initiation Request - Testing Ticket

The Technical Lead needs to send the Test Initiation Request to Testing Vertical Lead by creating a Testing Ticket in GELJIRA as per the format at Annexure-I along with Functional Requirement Specification(FRS) document or Change Request Form(CRF) whichever applicable towards a development request or change request ticket in GELJIRA.

3. Standard timeline for Testing team to test and find issues.

Application Form	Simple Form with grid and dependencies	8 hrs
	Average Form with grid and dependencies	12 hrs
	Complex Form with grid and dependencies	16 hrs
Application Flow	Simple Flow with hierarchy with reports	12 hrs
	Average Flow with report, payment, calculations and dependencies	18 hrs
	Complex flow with report, payment, calculations and dependencies	24 hrs

**4. Testing Compliance Timeline Standard for Developer to revert on testing issues**

	Low Priority	Normal Priority	High Priority
Simple Form	12 hrs	8 hrs	4 hrs
Average Form	20 hrs	12 hrs	8 hrs
Complex Form	24 hrs	16 hrs	12 hrs
Simple Flow	16 hrs	12 hrs	8 hrs
Average Flow	32 hrs	20 hrs	12 hrs
Complex Flow	48 hrs	32 hrs	16 hrs

Note:

1. If Compliance Timeline exceeds, than it will be reviewed by CTO/System Architect.
2. For high priority tasks (where compliance is required within 2 working days), if there is any deviation from the timeline due to **changes in requirements, addition of new requirements, shift in priority due to other critical tasks or any other justifiable reason**, the Project Manager must mandatorily provide a comment at the time the testing ticket is marked for retesting by Team Lead. This comment should clearly state the reason for the delay, ensuring accountability and transparency in the delivery process.

5. Test Applicability Standards

Below table gives a mapping of what tests are to be carried out for each type of service/code:

Sr No	Testing Types	Application Form	Dept Flow	Website	Windows	Mobile App
1	Smoke Testing	✓	✓	✓	✓	✓
2	Unit Testing	✓	✓	✓	✓	✓
3	Integration Testing		✓	✓	✓	✓
4	Responsive UI Testing	✓	✓	✓		
5	Performance and Load Testing as per requirement	✓		✓		✓
6	Security Testing	✓	✓	✓		✓
7	Regression Testing	✓	✓	✓	✓	✓
8	SEO Check list			✓		

Note: Sequence in which each test to be conducted is as per Sr. no mentioned starting from Smoke Test to SEO Checklist. Refer Annexure-II for information about testing types.



6. Standard Testing Tools

Testing Tool/Methodology						
Testing Type	Testim.io	Jmeter	BurpSuite	ZAP	SEO Review	Manual
Smoke Testing	✓					✓
Unit Testing (Functional, Validation and User Interface)	✓					✓
Integration Testing	✓					✓
Performance and Load Testing		✓				
Security Testing			✓	✓		
Regression Testing	✓					✓
SEO Check List					✓	

7. Sensitive personal data and information Security testing

Sensitive personal data/ Information of a person is data which cannot be shown to a third party or public access. The personal information consists of information relating to:

- 7.1. Password, Aadhaar card, Pan card.
- 7.2. Financial information such as Bank account or credit card or debit card or other payment instrument details.
- 7.3. Physical, physiological and mental health condition.
- 7.4. Sexual Orientation.
- 7.5. Medical records and history
- 7.6. Biometric information
- 7.7. Any detail relating to the above clauses as provided to body corporate for providing service; and
any of the information received under above clauses by body corporate for processing, stored or processed under lawful contract or otherwise



8. Testing Phase Standards

Each round of testing is treated as one single phase. The Testing should be cleared in max 4 phases(i.e Max 3 Compliances). Any deviation to this will have to be clarified and approved by System Architect. Technical Lead should provide detailed reason for deviation. Same details will have to be mentioned in Testing Clearance Certificate issued by Testing Vertical lead.

Phase I	Phase II	Phase III	Phase IV	more than Phase IV
✓	✓	✓	✓	

Note: The Project Manager or Technical Lead is authorized to mark a Testing Ticket as "On Hold" under the following conditions:

- Pending development work
- Unresolved dependencies
- Blocking issues affecting testing progress
- Any other valid reason delaying the testing process

A brief comment stating the specific reason for marking the ticket as "On Hold" must be added by Project Manager or Technical Lead at the time of status update to ensure clarity and traceability.

9. Testing Clearance Certificate - Refer Annexure III

- After completion of testing round of each module when no issues/bugs are found, Test Lead prepares a Testing Clearance certificate towards each module which includes Project and Testing task details, Types of testing performed, testing phase summary, Exception issues counts and details, and Approval Signatures of Test Lead and CTO. Test Lead submits the Certificate to CTO for approval. Once approved, Test Lead uploads the clearance certificate in the Testing Ticket in GEL JIRA and marks the ticket as 'Testing Cleared' with request to raise ticket for Integration testing.
- Technical lead will raise a sub ticket for Integration testing towards the testing ticket raised.
- Once Integration testing is cleared with approval from Test Lead and CTO. Test Lead will mention remark to provide the service/application for smoke testing upon moving to production server.
- Testing team will perform end-to-end test towards each application type/category to give a final go ahead on working of the system.



10. Standard Unit Test Scenarios for any application/ website.

A test scenario is a specific condition or situation that a tester creates to validate a particular aspect or functionality of a software application. It represents a high-level description of a test case or a set of related test cases that focus on testing a specific feature or requirement.

10.1 For Each application there will be many test scenarios which can be created and tested as per functionality of the application.

10.2 Each Applications which comes for testing has some common testing scenarios which can be tested before giving application for testing by developers end which can lead to less bugs in initial phases.

10.3 Smoke Testing Scenarios - Refer Annexure IV

Smoke testing scenarios ensure that most important functions of a program work correctly. It helps to identify major issues and defects early in the development process so that they can be easily fixed before proceeding with more comprehensive and detailed testing.

Smoke testing is done manually as well test automation tool **Testim.io** is used .

10.3 Unit Testing Scenarios - Refer Annexure IV

Unit test scenarios ensures that each unit of the software performs as expected. By focusing on small, manageable parts of the application, unit testing helps identify and fix bugs early in the development process, significantly improving code quality and reliability.

Unit test scenarios include Functional Test scenarios , Validation Test scenarios and User Interface Test scenarios

Functional Test scenarios

Functional test scenarios ensures that each application feature works as per the software requirements. Each function is compared to the corresponding requirement to ascertain whether its output is consistent with the end user's expectations.

Validation Test scenarios

Validation test scenarios ensures that each field and functionality is validated as per the requirements and standards



User Interface Test scenarios

User Interface test scenarios focusses on checking the appearance, functionality and usability of the application in test as per the company standards

10.4 Integration Test Scenario - Refer Annexure V

Integration test scenarios include scenarios in which the different units, modules or components of a software application are tested as a combined entity. However, these modules may be coded by different programmers.

The aim of integration testing is to test the interfaces between the modules and expose any defects that may arise when these components are integrated and need to interact with each other. Once all the components or modules are working independently, then we need to check the data flow between the dependent modules

10.5 Regression Test Scenarios - Refer Annexure IV

Regression test scenarios ensure that changes or updates to an existing software product do not affect previously functioning features. This test scenarios should be done when bug fixes are done or software enhancements are done in the system. Regression test scenarios are same as functional , validation and user interface test scenarios

11. Standard Non Functional Test Scenarios for any application/ website.

11.1 Performance/Load Test Scenario - Refer Annexure VI

Performance/Load testing checks how systems function under the projected number of concurrent virtual users performing transactions over a certain period of time. In other words, load testing measures how systems handle expected load volumes. There are a few types of open-source load testing tools, JMeter being the most popular.

For Performance testing **GTMetrix** and **Pagespeed** Insights websites are used. For Load testing **JMeter** tool is used.

11.2 Security Testing - Refer Annexure VII

Security testing is an important aspect of software testing focused on identifying and addressing security vulnerabilities in a software application. It aims to ensure that the software is secure from malicious attacks, unauthorized access, and data breaches.



Security testing involves verifying the software's compliance with security standards, evaluating the security features and mechanisms, and conducting penetration tests to identify weaknesses and vulnerabilities that might be exploited by malicious actors.

a) The Testing Team will conduct Vulnerability Assessments (VA) at the beginning of each quarter (January, April, July, and October) for all active projects. The report to be submitted to CTO of security tests carried out on whichever projects.

b) Every project must pass a mandatory internal security audit before an external audit, ensuring all vulnerabilities are resolved and strengthening overall security.

c) For Security testing OWASP ZAP, Burp Suite and Acunetix testing tools are used.

11.3 SEO Testing - Refer Annexure VIII

An SEO test is an evaluation of how well a page is optimized for search engines. These tests, often performed with the help of software tools, evaluate various elements of a page to determine what areas are well-optimized and what requires improvement.

12. Validation Field Matrix- Refer Annexure IV

- Validation Field Matrix is defined by our organization to have uniform implementation of fields and constraints.

- The field matrix depicts for a specific field what type of control to be used and what validations should be done on the same and what alerts to be shown during application development.

- Each of these are explained below:

- a) Control Type
- b) Validations
- c) Alerts

12.1 Control Type

What control should be used to depict the field on the form is defined by this column.

The field can be any of this:

- a) Text field
- b) Drop down
- c) Numeric Field
- d) Special Character Field
- e) Alphabet Field
- f) Calendar for Date Field



g) Button

12.2 Validations

The validations or checks which are to be applied towards a specific control and field are as below:

- a) Mandatory(required) - Compulsory field.
- b) Length Restrictions - Data length of the field can allow the user to enter is mentioned by this constraint.
- c) Max Length & Min length -For ex: Password field min and max length are applicable in use case where the password need to be between 8 to 25 characters.
- d) Format - For Some fields there are predefined format which should be taken in consideration and data should be accepted in application as per format. For ex: PAN no, GSTN, Date etc.
- e) Validation Alerts - As per Matrix fields which are ticked marked for that validation alert should fire when invalid data or required data is not entered.

12.3 Alerts

- a) Successful message : Should be displayed once Data is saved successfully for eg: on final save button once application is submitted.
- b) Confirmation alerts : Should be displayed before carrying out any functionality on click of that particular button for ex. on click of delete button confirmation alert should be displayed.



Annexure-I

Testing Ticket needs to be submitted with following details in GelJira.

1. Link to access the module
2. Point of Contact (Developer)
3. Test Priority (High/Medium/Low)
4. Reason in case of high priority
5. Login Credentials
6. FRS Document *
7. Calculation, if any
8. Report Formats
9. Functionality change document, if any changes occurred in between phases
10. Test Data, if required to cross check
- Incase of Website
11. Document to cross check correctness of data
12. For Load Testing, required Time and Users
13. Functionality document required, if any

**Annexure-II**

Testing Type		Description
✓	Smoke Testing	A quick test to be carried out to know whether the critical features of the system are working or not and also check Broken links.
✓	Functional Testing	Validates the software system against the functional requirements/specifications. The purpose of Functional tests is to test each function of the software application, by providing appropriate input, verifying the output against the Functional requirements.
✓	Validation Related Testing	Validation Related testing is a type of testing which involves checking of the fields against valid and invalid data in accordance with the validations laid out in the Standard Validation Matrix or as per the requirement provided by the developer in the FRS.
✓	User Interface Testing	UI Testing considers the look, the feel and ease-of-use of an application.
✓	Integration Testing	Software modules are integrated logically and tested as a group. A typical software project consists of multiple software modules, coded by different programmers. The purpose of this level of testing is to expose defects in the interaction between these software modules when they are integrated.
✓	Responsive UI Testing	Responsive UI Checks the following scenarios: Whether alignment is missing, text is getting trimmed, calendar control is getting cut. Overall color scheme/theme of the device, Progress indicators when pages are loading, overall responsiveness of applications on multiple devices.
✓	Performance and Load Testing	Performance testing measures response times, transaction rates and other time sensitive requirements . The goal of performance testing is to verify and validate if the performance requirements have been achieved.



✓	OWASP Security Testing	The standard used for web testing is OWASP which include techniques like injection, Broken Authentication and Session Management, Cross-site scripting, Insecure Direct Object References, Security Mis-configuration, Sensitive Data Exposure, Missing Function Level Access Control, Cross-site Request forgery(CSRF), using components with known vulnerabilities, Un-validated Redirects and Forwards.
✓	Regression Testing	Regression testing is performed to confirm that a recent program or code change has not adversely affected existing features. Regression Testing is nothing but a full or partial selection of already executed test cases which are re-executed to ensure existing functionalities work fine.
✓	SEO Check list	✓ HTML Headings Checker ✓ Canonical URL / Location Checker ✓ Back Link Checker ✓ Mobile Friendly ✓ Page Speed Insight ✓ Duplicate Content Checker ✓ AMP Checker ✓ Internal Link Analyzer ✓ Bulk meta description checker ✓ Bulk Title tag checker ✓ Bulk meta robots checker ✓ Featured Snippet Tool ✓ Inline CSS Checker ✓ Favicon Checker ✓ Caching ✓ Alternative Text Tag ✓ Responsive Design Meta Tag ✓ Alternate Image text ✓ Plain text emails and web reference ✓ Sitemap Checker



Annexure III

Application Testing Clearance Document

Project Name :
 Testing Ticket ID :
 Testing URL :
 Testing Ticket Summary :
 Testing Ticket Created Date :
 Reporter :

Test Applicability Standard Checklist for Application Form

Sr No.	Testing Type	Conducted? (Yes/No)	Reason for not conducting
1	Smoke Testing		
2	Functional Testing		
3	Validation Related Testing		
4	User Interface Testing		
5	Responsive UI Testing		
6	Regression Testing		
7	Integration Testing		

Testing Phase Summary

Phase No	Testing Reported Date	Compliance Received Date	Total Issues Reported	Issue Subtype				Repeated Issues	Testing Performed by
				Functional	Validation	UI	Report		
I									
II									
III									

Exceptions

Issue Status	Count	Copy Attached?
Not an Issue*		
Closed (Issues ignored)*		
Done (Suggestions ignored)*		

Testing Efforts

Development Efforts (Hrs)	Testing Efforts (Hrs)	Testing Compliance Efforts (Hrs)

Testing Clearance Details

Phase No : Date :

Reason for excess Number of Phases : Attached Below

Reason for excess efforts spent in Testing : Attached Below

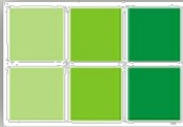
Reason for excess efforts in Testing Compliance : Attached Below

Remarks by Test Lead : Note: Functionality ready to go live.

Test Lead Sign with :
Date

CTO Comments & Sign :





Issues which are marked Closed and commented

Issues()	Issue by Tester	Date	Tester Name	Comment by Developer

Issues which are marked Not an Issue and commented

Issues()	Issue by Tester	Date	Tester Name	Comment by Developer	PM comment

Suggestions which are ignored , marked as Done and commented

Issues()	Issue by Tester	Date	Tester Name	Comment by Developer	PM comment

Reason of exceeding Phase IV of Testing

Commented By:

Technical Lead

Reason of excess efforts spent in Testing

Commented By:

Test Lead

Reason of excess efforts spent in Testing Compliance

Commented By:

Technical Lead

**Annexure IV**

Smoke Testing Scenarios	
✓	Check if without entering login credentials on click of login button , required alert gets fired for the fields
✓	Check if when without entering data in login fields or invalid data is entered on click of login button, Captcha gets regenerated
✓	Check if without entering Captcha , on click of login button , required alert fires for Captcha field and the Captcha refreshes
✓	Check if validation alert gets fired on click of login button when username is entered incorrect and password is entered correct
✓	Check if validation alert gets fired on click of login button when username is entered correct and password is entered incorrect
✓	Check if a user can log in with valid credentials.
✓	Check if without entering data in mandatory fields, on click of Save button , validation alert fires for mandatory blank fields
✓	Check if on entering invalid data in all the fields , on click of Save button , page does not redirect to error page
✓	Check if on entering max data in all the fields, on click of Save button , page does not redirect to error page and form gets saved
✓	Check if on multiple click of Save button , multiple applications do not get saved
✓	If there is draft save functionality , check if on entering valid data and on click of Save functionality form, gets saved and is displayed in the search grid
✓	Check if on click of edit towards the draft saved functionality , form opens in edit mode and is allowed to edit
✓	Check if draft saved application gets saved on changing the data and on click of Save button
✓	Check if there is delete button towards the draft saved application
✓	Check if file gets uploaded on selecting valid file and clicked on upload button
✓	Check if on click of View towards uploaded document , document should be correctly displayed in the document viewer
✓	Check if on click of delete towards uploaded document , document gets deleted
✓	Check if on saving form details and uploading documents , on click of final Save / Submit button , the form gets submitted and is displayed in the search grid
✓	Check if on click of links or urls , page redirects to the correct target page
✓	Check if application status at all the stages of process is correctly displayed
✓	Check if on entering valid search criteria , entries related to the search criteria get populated in the grid correctly

**Annexure V**

Unit Testing Test Scenarios	
Functional Test Scenarios	
✓	Check that Drop-down fields have the first entry as blank or text like 'Select' or 'Choose One'
✓	Check that on click of Any button or link, the action of the same should be correctly performed as required instead of displaying any error or redirecting to any wrong page.
✓	Check that once entry is submitted and then on refreshing the page the form or entry should not be allowed to submit again.
✓	Check that every text field should have data limit set for it and on entering max data in fields, page should not get redirected to error page on entering or on click of any proceed/save button.
✓	Check that on multiple click on Save/Submit/Add button, entry should not get saved/submitted/added multiple times.
✓	Check that when any entry is saved as draft, it should be shown with delete button.
✓	Check Calculation if any, should be checked and should work as per the functionality.
✓	Check that if downloading any Report/File, correct and desired file should get downloaded.
✓	Check that Non-mandatory date fields should have a cross to clear the date field.
✓	Check that when one entry is in edit mode, other entry should not be allowed to add/edit/delete, validation alert should be displayed to save the entry in edit mode.
✓	Check Correctness of Data: In Edit mode or in View mode, the saved/submitted data should get correctly populated and displayed as saved/submitted.
✓	Check that when non mandatory fields are kept blanked, in edit mode it should not be displayed as NA or null.
✓	Check that on multiple click on Save/Submit/Add button, entry should not get saved/submitted/added multiple times.
✓	Check that user should not be able to submit page twice by pressing submit button in quick succession.
✓	Check that after entering/selecting all the mandatory data, on click of Save/Submit/Add button, data should get saved/submitted/added properly.
✓	Check that on click of the correct menu or on entering proper URL, correct page should be redirected and shown to the user.
✓	Check that on deleting the entry, the entry should get deleted and on refreshing or re opening the page, the deleted entry should not reappear.
✓	Check that Reset button functionality should set default values for all fields.
✓	Check that Select/deselect all records option should be provided if page supports record add/delete/update functionality.
✓	Check page sorting functionality after add/edit/delete operations on any record.
✓	Check if downloadable files are pointing to the correct file paths.



✓	Check that on application crash or unavailable pages should be redirected to the proper error page.
✓	Check the dropdown option by selecting and then unselecting, page should not redirect to error page.
✓	Check that for pagination, when clicked on last page number and then clicked on any other page/first page , it should not redirect to error page. Also, when number is clicked multiple times in quick succession, it should not redirect to Error page.
✓	Check that Password and other sensitive fields should be masked while typing.
✓	Check that scroll bar should be enabled only when necessary.
✓	Check that user should not be able to type in drop down select lists.
✓	Check that information filled by users should remain intact when there is an error message on page submit. The user should be able to submit the form again by correcting the errors.
✓	Check that page loading symbol should be displayed when it's taking more than default time to load the result page.
✓	Check that result grids should include all the specified columns with correct values based on the search criteria entered
✓	Check that pagination should be enabled when there are more results than the default result count per page.
✓	Check for Next, Previous, First and Last page pagination functionality.
✓	Check that duplicate records should not be displayed in the result grid.
✓	Check that on click of entry in grid it should redirect to specific page.
✓	Check for uploaded image path.
✓	Check image upload and change functionality.
✓	Check image upload functionality with image files of different extensions (e.g. JPEG, PNG, BMP etc.).
✓	Check during file upload, progress bar should appear for large size files.
✓	Check if cancel button functionality is working in between upload process.
✓	Check multiple file upload functionality if same document type or same document name are not allowed as per functionality
✓	Check image quality after upload. Image quality should not be changed after upload.
✓	Check if the user is able to use/view the uploaded files.
✓	Check for Delete functionality for uploaded file.
✓	Check for download functionality for uploaded file.
✓	Check for Malicious file upload. System should not allow malicious file and proper alert message should be shown on uploading invalid file.
✓	Check if the correct fields are highlighted in case of errors.



✓	Flow: It should be ensured that properly flow is maintained, on submitting/ verifying/ complying / forwarding/ approving/ rejecting/ reverting, Entry should get marked to the desired user as per the functionality requirement.
Validation Test Scenarios	
✓	Check that all mandatory fields should be validated and indicated by an asterisk (*) symbol.
✓	Check that validation error messages should be displayed properly at a correct position.
✓	Check that without selecting mandatory data on click of Save/Submit/Add button, required alert should be displayed for mandatory fields.
✓	Check numeric input fields with character input values. Proper validation message should appear.
✓	Check that on entering invalid data in fields, on click of Save/Submit/Add button, entry should not be allowed to save/submit/add, and proper validation text message should be shown to the user. No invalid data should be allowed to get saved/submitted.
✓	Check that all error messages should be displayed in the same CSS style (e.g. using red colour).
✓	Check that general confirmation messages should be displayed using CSS style other than error messages style (e.g. using green colour).
✓	Check that Hint messages / Tool tips text should be meaningful.
✓	Check that Input fields should be checked for the max field value. Input values greater than specified max limit should not be accepted or stored in the database.
✓	Check if numeric fields accept negative , exponential values
✓	Check that if numeric fields do not accept decimals , check by entering decimal values
✓	Check that if numeric fields accept decimals values, check by entering more than 2 digits after decimals . Proper validation alert should get fired
✓	Check amount fields with decimal number values. Amount fields should accept only 2 digits after decimals and proper validation alert should get fired for invalid data
✓	Check for Area field validation . Area fields should accept only 3 digits after decimals and proper validation alert should get fired
✓	Check all input fields for special characters.



✓	Check that Delete functionality for any record on a page should ask for a confirmation message.
✓	Check that for duplicate entry in the grid/table, validation alert should be displayed 'Entry already exist'.
✓	Check From date and To date validation, From date should be always be earlier than To date.
✓	Check that confirmation messages should be displayed before performing any update, delete, submit operation.
✓	Check document upload functionality with documents having space or any other allowed special character in the file name than proper error message should be displayed
✓	Check document upload with document size greater or less than the allowed size. The Proper error message should be displayed.
✓	Check document upload functionality with file types other than the allowed file types (e.g. txt, doc, pdf, exe etc.). A Proper error message should be displayed.
✓	Document Upload: Proper note should be shown mentioning the acceptable file types and acceptable min and max size.
User Interface Test Scenarios	
✓	Check that all fields on a page (e.g. text box, radio options, drop-down lists) should be aligned properly.
✓	Check that Font size, style, and colour for headline, description text, labels, infield data, and grid info should be standard as specified in FRS.
✓	Check that Numeric values should be justified correctly unless specified otherwise.
✓	Check that amount values should be displayed with correct currency symbols.
✓	Check that Field labels should be standard e.g. field accepting user's first name should be labeled properly as 'First Name'.
✓	Check text on all pages for spelling and grammatical errors.
✓	Check that enough space should be provided between field labels, columns, rows, error messages etc.
✓	Check if proper field labels are used in error messages.
✓	Check if the drop-down list options are readable and not truncated due to field size limit.
✓	Check that Result grids should be displayed with proper column and row spacing.
✓	Check if images of specified height and width (if defined) are accepted otherwise rejected.
✓	Check that while uploading of image check its orientation of image to ensure image to appear proper.

**Annexure VI**

Integration Testing Test Scenario	
✓	Verify that data passed from Module A to Module B is correctly received and processed.
✓	Ensure that Module B can handle erroneous or unexpected data from Module A.
✓	Check that the interfaces (APIs, function calls) between the two modules are compatible.
✓	Verify that state changes in Module A affect Module B as expected.
✓	Ensure that the sequence of operations between the two modules is maintained
✓	Verify that data exchanged between the two modules is secure
✓	Test how both modules handle boundary conditions and limits.
✓	Ensure that both modules work correctly with different configurations or settings
✓	Verify that different user roles (e.g., employee, manager, HR) can log in and access only the functionalities relevant to their role.
✓	Ensure that unauthorized access to restricted sections is prevented.
✓	Verify that unique Acknowledgment no.is generated and this acknowledgment no.is reflected across all related modules.
✓	Verify that when a request is raised, its status is correctly updated and displayed towards the applications.
✓	Check if any change in request status (e.g., approved, rejected) is updated across all relevant views.
✓	Ensure that edited changes at any stage by any users are properly reflected in the system with the same unique ID.
✓	Ensure that data entered in one module (e.g., request form) is consistently displayed in other modules (e.g., records, reports) without discrepancies.
✓	Confirm that email notifications are triggered when actions are taken (e.g., request approval, rejection, or sending back with comments).
✓	Ensure that notifications are sent to the correct recipients and the content reflects the action taken.
✓	Ensure no data loss or corruption during multi-user interactions.
✓	Session Management - Test session management by ensuring that users are logged out after a period of inactivity and that pending actions are preserved across sessions.
✓	Verify that any actions taken right before session expiry (e.g., submitting a form) are saved correctly after login.
✓	Concurrency Handling - Verify that if multiple users are interacting with the same request simultaneously the system handles concurrent updates without data conflicts.
✓	Validation Across Modules - Test that all necessary validations (e.g., required fields, file format checks) are applied consistently across modules and reflected correctly in error messages.
✓	Third-Party Integration - Verify that third-party services (e.g., email notifications, external validation tools) are properly integrated and triggered at the right points in the workflow.
✓	Pagination and Data Retention - Check that when pagination is enabled (e.g., in records or task tabs), selections, filters, and data retention are maintained when navigating between pages.



✓	Error Handling and Recovery - Ensure that errors (e.g., network interruptions, invalid data submissions) are handled gracefully, with proper feedback to the user and without data loss.
✓	File Upload and Validation Across Modules -Verify that when files (e.g., supporting documents) are uploaded in one module, they are accessible and validated correctly in all related modules.
✓	Ensure file type and size validation is consistent across different modules where file uploads are required.
✓	Check that application processed at particular stages of the workflow , are displayed in the correct buckets in the user's dashboard.
✓	Cross-Browser and Cross-Platform Compatibility - Test the system's behavior across multiple browsers (e.g., Chrome, Firefox, Edge) and platforms (e.g., Windows, macOS, mobile) to ensure that integration and data flow remain consistent.
✓	Bulk Data Handling Across Modules - Ensure that each request in the bulk operation (e.g., bulk approval or rejection of requests) is updated correctly in records, task lists, and notifications
✓	Dependent Dropdowns and Data Binding - Test that dependent dropdowns (e.g., country and state selection) are correctly bound across modules. Changing one value should update the related fields throughout the system.
✓	Test that users are able to pick up workflows from where they left off, even after switching between modules.
✓	Ensure that invalid data rejected in one module cannot be accepted when entered into another module.

**Annexure VII**

Performance/Load Testing Test Scenarios	
✓	Check if the page load time is within the acceptable range
✓	Check the page load on slow connections.
✓	Check the response time for any action under a light, normal, moderate, and heavy load conditions
✓	Check CPU and memory usage under peak load condition.
✓	Check if the link under test runs under default load conditions (1 user - 1 sec)
✓	Measure the response time under normal load conditions and ensure it meets the acceptable threshold (e.g., less than 3 seconds for each action)
✓	Simulate multiple users (e.g., 100, 500, 1000 users) accessing the system simultaneously to check if the system can handle the load without significant performance degradation.
✓	If the test runs successfully in default settings , increase the number of users . If the test fails in default condition , than increase the ramp-up time and again run the test.
✓	Check increasing the number of users and ramp up time in different combination based on if the test is pass or failed.
✓	Monitor and measure the system's response time, CPU usage, and memory utilization during the load.
✓	Observe how the system behaves when overloaded and ensure it degrades gracefully, providing informative error messages instead of crashing.
✓	Gradually increase the number of users or transactions (e.g., from 100 to 1000 users) to check how well the system scales.
✓	Ensure that file transfer speeds remain consistent and the system does not slow down or time out during heavy file handling.
✓	Test how quickly the system responds to user actions (e.g., form submissions, page loads, API calls)
✓	For performance testing , enter the url of the website to be checked on GTMetrix and PageInsight website and check for the results generated

**Annexure VIII**

Security Testing Test Scenarios	
✓	Check for SQL injection attacks.
✓	Secure pages should use HTTPS protocol.
✓	Page crash should not reveal application or server info. Error page should be displayed for this. Error messages should not reveal any sensitive information.
✓	All credentials should be transferred over an encrypted channel.
✓	Check application logout functionality.
✓	Check for Brute Force Attacks.
✓	Check session cookie duration and session termination after timeout or logout.
✓	Session tokens should be transmitted over a secured channel.
✓	The password should not be stored in cookies.
✓	Sensitive fields like passwords and credit card information should not have to autocomplete enabled.
✓	File upload functionality should use file type restrictions and also anti-virus for scanning uploaded files.
✓	Password and other sensitive fields should be masked while typing.
✓	Check if access privileges are implemented correctly
✓	Verify CAPTCHA functionality
✓	Test for Cross-Site Scripting (XSS) vulnerabilities (stored, reflected, DOM-based).
✓	Ensure that Cross-Site Request Forgery (CSRF) protections are implemented.
✓	Validate that security headers (CSP, X-Frame-Options, etc.) are present and correctly configured.
✓	Ensure that sensitive data is encrypted both in transit and at rest.
✓	Check for proper handling of sensitive data in logs and error messages.
✓	Verify that third-party libraries are up to date and free of known vulnerabilities.



✓	Test for information disclosure through unnecessary HTTP methods (OPTIONS, TRACE).
✓	Check for Denial of Service (DoS) vulnerabilities by simulating high load.
✓	Validate session management, ensuring users are logged out after a period of inactivity.
✓	Ensure that application state cannot be manipulated through URL tampering.
✓	Check for proper permissions on files and directories.
✓	Test for security of API endpoints, validating authentication and authorization mechanisms.
✓	Ensure that sensitive information is not exposed through URL parameters.
✓	Review code for hardcoded secrets, such as API keys or passwords.
✓	Check for directory traversal vulnerabilities that allow access to restricted files.
✓	Validate that user input is sanitized and validated to prevent injection attacks.
✓	Verify proper use of secure coding practices and frameworks throughout the codebase.
✓	Test the application under different user roles to verify access control.
✓	Validate that all forms of data transfer (APIs, file uploads) are secure and validated.
✓	Ensure that logout functionality terminates all active sessions.
✓	Verify that password reset mechanisms are secure and do not leak information.
✓	Check that sensitive data is not stored in local storage or session storage in web applications.
✓	Test for improper implementation of SSL/TLS (including weak cipher suites).
✓	Verify that application code does not expose sensitive configurations or secrets.
✓	Test for vulnerability to Clickjacking attacks.
✓	Check for information leakage through headers or comments in the source code.
✓	Test for race conditions that could lead to security issues.
✓	Validate proper handling of expired or revoked tokens in authentication flows.
✓	Ensure that all endpoints require authentication and authorization.
✓	Validate the implementation of rate limiting on APIs to prevent abuse.



✓	Verify that sensitive operations require re-authentication or additional verification.
✓	Ensure that proper timeout and inactivity handling is implemented for sessions.
✓	Test for vulnerabilities related to insecure direct object references (IDOR).
✓	Validate that the application correctly implements secure coding standards.
✓	Check for missing or incorrect HTTP security headers (HSTS, Content-Security-Policy).
✓	Test for the presence of unnecessary HTTP methods (PUT, DELETE) on public endpoints.
✓	Ensure that all error codes and messages are generic and do not leak system information.
✓	Check for vulnerabilities related to improper caching of sensitive data in browsers.
✓	Validate that API responses do not expose sensitive data or system information.
✓	Ensure proper input length validation to prevent buffer overflow attacks.
✓	Test for insecure usage of cryptographic functions (e.g., weak algorithms).
✓	Test for vulnerabilities in the configuration of web servers (e.g., Apache, Nginx).
✓	Validate that all temporary files created by the application are securely handled.
✓	Check for misconfigured CORS policies that could lead to security vulnerabilities.
✓	Validate that all cryptographic keys are managed securely (e.g., rotated, protected).
✓	Ensure that all forms of user input are properly validated and sanitized server-side.
✓	Check for potential vulnerabilities in JSON Web Tokens (JWT) handling and validation.
✓	Ensure that no sensitive information is included in URLs or query parameters.
✓	Test for exposure of sensitive data in response headers.
✓	Validate the security of the application's configuration files (e.g., .env files).
✓	Ensure that application dependencies are scanned regularly for vulnerabilities.
✓	Check for session fixation vulnerabilities during the login process.
✓	Test the effectiveness of security patches and updates in the application.

**Annexure IX**

SEO Testing Test Scenarios	
✓	Ensure that all pages have appropriate meta titles and descriptions.
✓	Verify the correct use of header tags (<h1>, <h2>, etc.) for content structure.
✓	Check for clean, descriptive, and keyword-rich URLs
✓	Validate that images have appropriate alt attributes
✓	Confirm the presence and accessibility of an XML sitemap.
✓	Validate the configuration of the robots.txt file
✓	Ensure that the website is mobile-friendly
✓	Check the loading speed of pages
✓	Verify that internal links are functioning and logically structured.
✓	Assess the quality and relevance of external links.
✓	Confirm the presence of canonical tags to avoid duplicate content issues.
✓	Verify the implementation of structured data
✓	Ensure custom 404 error pages provide helpful information
✓	Verify that the website uses HTTPS.
✓	Assess the quality of content and its alignment with target keywords.
✓	Check for proper integration of social sharing features
✓	Verify that tracking codes are correctly implemented.
✓	Check for local SEO optimizations if applicable



Annexure X Validation Field Matrix

	Control Type							Validations					Alerts				
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calendar	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
First Name	✓			✓	✓			✓	50	✓			✓			Allow only [A-Z], [a-z], dot(.), apostrophe('), @ , space	
Middle Name	✓			✓	✓				50	✓			✓			Allow only [A-Z], [a-z], dot(.), apostrophe('), @ , space	
Last Name	✓			✓	✓			✓	50	✓			✓			Allow only [A-Z], [a-z], dot(.), apostrophe('), @ , space	
Business Entity Name	✓		✓	✓	✓			✓	150	✓			✓			Allow only [A-Z], [a-z], [0-9], dot(.), apostrophe('), @ , &, hyphen(-), space, /, ()	
Email ID	✓		✓	✓	✓			✓	160	✓		✓	✓			Allow [A-Z], [a-z],[0-9], special characters in mail id format ex: abc@xyz.com / abc@xyz.co.in	Should not accept @ twice and also dot (.) should not be mention more than once together.
Website	✓		✓	✓	✓				160	✓		✓	✓			Allow only [A-Z], [a-z],[0-9], hyphen (-) , dot(.) in format ex: https://www.abc.com or https://abc.com or http://abc.com	
Password	✓		✓	✓	✓			✓	25	✓	✓		✓			Allow only [A-Z], [a-z],[0-9] and special characters.	Password should accept more than 8 characters and Maximum character limit is 25 with combination of Alphabets , numbers and special characters.Password should be encrypted and hashed at client side before passing to server



	Control Type							Validations					Alerts				
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calendar	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
PAN Card No	✓		✓		✓			✓	10			✓	✓			Allowed only [A-Z],[a-z],[0-9]	ABCDE1234F [first 5 alphabets, 4 numbers , last 1 alphabet] and While storing store in encrypted format in db
Aadhaar Number	✓		✓					✓	12	✓		✓	✓			Allowed only [0-9]	Aadhaar no. Accept 12 digit; Cannot start with 0 & 1, While storing store in encrypted format in db; while viewing show last 4 digit only; Before using the aadhaar number department should notify usage in the gazette ; Should be encrypted and hashed at client side before passing to server. 2. Aadhaar consent should be mandatory
Virtual ID	✓		✓						16	✓			✓			Allowed only [0-9]	Virtual ID Accept 16 digit.
MobileNumber	✓		✓					✓	10*	✓	✓	✓	✓			Allowed only [0-9]	*if business requirement demands international numbers then add dropdown for Country code before the field and allow max length 15
Landline Number	✓		✓						15	✓	✓		✓			Allowed only [0-9]	
Fax Number	✓		✓						15	✓	✓		✓			Allowed only [0-9]	



	Control Type							Validations					Alerts				
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calender	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
Address	✓		✓	✓	✓			✓	300	✓			✓			Allow only [A-Z], [a-z],[0-9], dot(.), apostrophe('), @ , & , space, open and close bracket () , [],underscore, (_) , comma(,) , dash(-) slash (/).	
HouseName/HNo.	✓		✓	✓	✓			✓	50	✓			✓			Allow only [A-Z], [a-z], [0-9], dot(.), apostrophe('), @ , & , space, open and close bracket () ,[], underscore, (_) , comma(,) , dash(-) slash (/).	
Locality	✓		✓	✓	✓			✓	250	✓			✓			Allow only [A-Z], [a-z], [0-9], dot(.), apostrophe('), @ , & , space, open and close bracket () ,[], underscore, (_) , comma(,) , dash(-) slash (/).	
City/Town/Village		✓			✓			✓	150	✓			✓			For Goan Cities/Villages use Label as City/Village and use dropdown; For other state cities use 150 char city/village name	
State		✓						✓					✓			Dropdown only for Indian States; By default Goa state to be selected	



	Control Type							Validations					Alerts				
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calender	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
District		✓						✓					✓			To be shown if needed; For Goa State districts can be auto handled on Taluka.. Should be dropdown	
Taluka		✓						✓					✓			Dropdown only for Indian States; By default Goa state talukas to be listed... for Goa state the key can be combination of District and taluka to avoid showing the district dropdown	
PinCode	✓		✓					✓	6	✓		✓	✓			Allowed only [0-9]	* ZIP code should be 6 digit
Date of Birth						✓	✓	✓	11	✓		✓	✓			Allowed only [0-9], [a-z],,[A-Z], hyphen(-)	Date Picker recommended and Check if business requirement has only specific age limit then add restrictions on years.Format dd-MMM-YYY
Date						✓	✓	✓	11	✓		✓	✓			Allowed only [0-9], [a-z],,[A-Z], hyphen(-)	Date Picker Recommended and check If using for from date & to date...check From date is smaller than to date... also check if business requirement has specific date range. Format dd-MMM-YYYY



	Control Type							Validations						Alerts			
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calendar	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
Year	✓	✓	✓					✓	4	✓		✓	✓			Allowed only [0-9] if Text Field.	text or dropdown as per business requirement and if Text field than Accept 4 digits.
Upload Documents							✓	✓	*	✓	✓	✓	✓	✓		Content Validation, File Type and length validation as per business requirement The front-end should show allowed file types, max size, client side validation of file length before uploading to avoid bandwidth consumption. Images to be compressed to min 96dpi for A4 size and 120dpi otherwise also the resolution to be corrected to have optimal image size.Orientation of image file to be checked and uploaded correctly.	1. Malicious file should not be allowed to upload. 2. When clicked on button or uploaded document link multiple time it should only able to view only once not multiple times. 3. Message alerts to be shown when image/Doc's are uploaded successful or upon error
Amount	✓		✓	✓				✓	15	✓			✓			Allowed only [0-9], dot(.)	Tempering of amount should not be allowed from developer tools on page nor it should be allowed from any security tool.
DD/Cheque Number	✓		✓		✓			✓	15	✓		✓	✓			Allowed only [0-9]	Should allow to accept numbers starting with '0'.



	Control Type						Validations					Alerts					
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calender	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
OTP	✓		✓					✓	6	✓		✓	✓	✓		It should not allow to submit previous OTP after creating new OTP.	OTP should have some time limit and within that time on click of regenerate button some alert should display saying to wait for mention amount of time.
GSTN Number	✓		✓		✓			✓	15	✓		✓	✓			Allowed only [A-Z],[a-z],[0-9]	GSTN number Accept 15 characters combination of Alphabets and Numbers ; Format ex: 30ABCDF1234F1Z2
Voter Id	✓		✓		✓			✓	10	✓		✓	✓			Allowed only [A-Z],[a-z],[0-9], /	Format first 3 letters,7 numbers eg. CPQ0640192
UAN Number	✓		✓						12	✓		✓	✓			Allowed only [0-9]	UAN No. Accept 12 digit.
Udyog Aadhaar number	✓		✓		✓				12	✓		✓	✓			Allowed only [A-Z],[a-z],[0-9]	Udyog Aadhaar number Accept 12 characters combination of Alphabets and Numbers ; Format ex: GA01A0000001
Bank		✓						✓									
Branch		✓						✓									
IFSC	✓		✓		✓			✓				✓				Allowed only [A-Z],[a-z],[0-9]	IFSC code should get Auto-populated as per selected Bank and Branch and should also be able to enter manually as per requirement.
Account Number	✓		✓		✓			✓	16	✓	✓		✓			Accept account number twice one masking and second plain text and compare both before allowing submission	Should not allow to copy paste and While storing store in encrypted format in db
Count	✓		✓							✓	✓		✓			Allowed only [0-9]	



	Control Type							Validations					Alerts				
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calender	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
Detail Section													✓			Always show the items list in a data-list/grid view compatible with responsive layout with "Add" button and then show screen for entering the details..	Check if business process requirement; If compulsory check if unsaved item in the fields and add/update button is not clicked which might loose data
Delete							✓							✓	✓		Date to be validated with server date for comparisons.
Save Button							✓							✓	✓	Multiple times entry should not get saved when clicked on multiple time on save button	If Max and multi-line data is entered in fields,Save button should not give error message or there should not be any issue while saving.
Submit Button							✓							✓	✓	Multiple times entry should not get saved when clicked on multiple time on submitted button	If Max and multi-line data is entered in fields,Submit button should not give error message or there should not be any issue while submitting.
Back Button							✓										if going to different page without saving changes
Next button							✓										if going to next page with/without saving changes. Depending on the business requirement where user is not allowed to move ahead without saving current page.
Previous Button							✓										if going to previous page without saving changes
Update Button							✓							✓	✓		While updating details new entry should not get added to the grid earlier details should get updated.
Add to Grid							✓								✓	Multiple times entry should not get added to grid when clicked on Add button multiple times.	



	Control Type						Validations					Alerts					
Fields	Text Fields	Dropdown	Numeric fields	Special character field	Alphabet fields	Calender	Button	Mandatory(required)	Length Restrictions	Max Length	Min Length	Format	Validation Alerts	Successful message	Confirmation alert	Additional Data Conditions	Remarks
Delete from Grid							✓							✓	✓		When one entry is in edit mode another entry from grid should not be allowed to delete.
Edit Grid Item							✓							✓	✓		When one entry is in edit mode another entry should not be allowed to open in edit mode.
Online Payment Amount	✓		✓	✓				✓		✓				✓	✓	Allowed only [0-9]	Amount field should be disable and it should not be able to modify from developer tool or any other security tool or if modified than action performed should not get reflected.